

Matthew Zhu

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Education

University of Waterloo

Graduating Aug 2029

BCS, Honours Computer Science with Co-op

GPA: 3.9/4.0

- **Relevant Coursework:** Object-Oriented Software Dev, Logic and Computation, Elementary Algorithm Design
- **Scholarships:** René Descartes National Scholarship Recipient (**1 of 10 in Canada**) - \$25,000 (2025)

Skills

Languages: Python, C/C++, Rust, Java, JavaScript, TypeScript, PHP

Frontend: React, Next.js, Node.js, HTML5, CSS3, Pygame, Hugo

Backend and Databases: Flask, FastAPI, REST APIs, PostgreSQL, SQLite, Firebase, Supabase, Pandas

AI/ML: PyTorch, Hugging Face, LangChain, RAG, Modal, CNNs, Data Preprocessing

Tools & Platforms: Linux, Git, GitHub, Bitbucket, Docker, Cloudflare, VSCode, Jira, LaTeX

Certifications: Stanford CS 230 Deep Learning, Google for Developers Machine Learning Course

Work Experience

Software Engineering Intern

Ottawa, ON

Ciena

May 2026 - Present

- Architected a **Rust** port of a **1800-line C** optical-planning model using **PyO3/mathurin** and **cbindgen**
- Built **7 neural network models** using **HuggingFace Candle**, replicating a **TensorFlow/ONNX** architecture
- Engineered a **FFI** bridge to expose a **Rust/Candle ML backend** as a library for a **C-based network simulator**
- Implemented a **ONNX/Candle inference mode**, validating zero divergence using a **500-case regression suite**
- Cross-compiled a **statically-linked Rust binary**, enabling **ML inference** on legacy **Linux/Docker** systems

Software Developer

Remote

Auxilium

Jun 2025 - Present

- Analyzed historical pairing outcomes to identify key mentor-student matching drivers using **OOP**
- Designed and implemented a **matching algorithm in Python (Pandas)** to rank mentor-student compatibility
- Optimized matching logic using weighted criteria and constraint filtering, increasing successful match rates by **36%**
- Built an indexed **SQLite relational database** to support efficient querying and repeatable matching runs

Full-Stack Development Experience

ClipFarm

[GitHub Link](#) 

Python, TypeScript, FastAPI, Next.js, Celery/Redis, PostgreSQL, PyTorch, Docker

Mar 2026 - Present

- Engineered a full-stack video pipeline with **Next.js** and **FastAPI** that turns volleyball VODs into highlight clips
- Designed **ball tracking (RF-DETR)** detecting rally contacts, refined by **YOLOv8-pose** classification
- Offloaded inference to **serverless GPUs** with CPU fallback, cutting per-game tracking from **13 to 8.7 minutes**
- Built a **Cloudflare R2** cache keyed by video hash, and skip-decode frame sampling for **31% faster** tracking
- Shipped async **Postgres (Supabase)** REST endpoints with **JWT auth** and a **Dockerized** multi-service stack

HUME

[GitHub Link](#) 

Python

Jul 2026

- Engineered a **structural firewall** for an online **belief-revision** engine to block **prompt-injection**
- Built a **log-odds** likelihood-ratio update that revises confidence from structured **provenance** rather than text
- Designed **structural OOD detection** to separate in-model contradictions from out-of-model regimes
- Isolated text parsing behind a **perception seam** with **NegEx/ConText**-style negation and modality scoping
- Developed a **19-probe adversarial** test battery (paraphrase, renamed-entity, injection) to measure generalization

Lumi

[GitHub Link](#) 

Python, PyTorch, FastAPI, PyMuPDF, ChromaDB, LangChain, RAG

Jan 2026 - Present

- Built a **FastAPI** backend with for mentor matching and math problems, integrating **LangChain** for task routing
- Engineered a **RAG system** using **ChromaDB** to search **500+** math contest problems with topic-aware filtering
- Developed an **ML mentor matcher** to score mentors by subject alignment and qualifications with **PyTorch**
- Implemented a **PDF ingestion pipeline** for contest problems using **PyMuPDF**

Achievements

- Canadian Chess **National Master** Title, peak rating of **2251** CFC and **2136** FIDE
- Represented **Team Canada** at the 2019 World Cadet Chess Championships
- **3x Top 3** in Ontario Chess Championships, **2x Top 5** in Canadian Chess Championships
- Represented the **University of Waterloo** at the 2026 Canadian University Chess Championships
- **Canadian Mathematical Olympiad** Repêchage Qualifier (**Top 100**), **2nd Place** in Eastern Ontario (2025)
- Competed for the Carleton Math Enrichment Centre at the **Harvard-MIT Mathematics Tournament** (2025)